# Revision Notes for Class 10 Science

## Chapter 5 - Life Processes

The study of living things is referred to as biology. Organisms refer to all living things, which includes both plants and animals. However, seven life processes influence how we determine if something is alive or not. If anything is alive, it will go through the 7 life processes listed below.

### 1. Movement

- Animals and plants both have the ability to move around. Plants are rooted and grow at a leisurely pace.
- Their roots go deeper into the soil, while their stems reach for the sun. Animals, on the other hand, can shift their entire bodies swiftly.
- They have the ability to migrate in pursuit of food, shelter, or safety.

### 2. Respiration

- The process of getting energy from the food we eat is known as respiration.
- All living things need to breathe in order to develop, replace worn-out parts and move. Respiration occurs in the cell's mitochondria.

### 3. Sensitivity

- All living things are sensitive, which means they are aware of changes in their surroundings.

- Heat, light, sound, touch, and chemicals with taste and smell are all stimuli that animals respond to swiftly.
- Plants, on the other hand, appear to be less sensitive and respond more slowly.

**4. Growth**

- Growing is a process that occurs in all living creatures. Plants keep growing for the rest of their lives.
- When animals reach adulthood, they stop growing.
- Even when development ceases, an animal's body continues to be replenished with materials obtained from its food.

**5. Excretion**

- All living things produce waste products, which can be either useless or hazardous to them and hence must be disposed of.
- The process of excretion is the removal of metabolic waste.
- Plants retain waste substances in their leaves, and as the leaves fall off, the garbage is eliminated.
- Other waste substances exit the body in urine and sweat, and animals breathe out waste carbon dioxide.

**Note:** Egestion, not excretion, is the process of getting rid of faeces or undigested food.

**6. Reproduction**

- In order for their species to survive, all living organisms must produce offspring who are genetically identical to themselves.
- Reproduction is the term for this process. Plants generate seeds, which germinate and produce new plants of the same species.
- Animals either lay eggs or have children. There are two types of reproduction: sexual and asexual.
- Sexual reproduction includes two parents and the union of two gametes, whereas asexual reproduction involves one parent reproducing itself.

**7. Nutrition**

Food is required for energy and growth in both plants and animals. Photosynthesis allows plants to produce their own food. Simple molecules like carbon dioxide and water are converted into more complex carbohydrate molecules using sunlight. Animals are unable to produce their own food and must rely on the nourishment of other plants and animals. Animals consume complicated substances and deconstruct them into small, soluble molecules that can be utilised for energy and growth.

- **Nutrition:** The process of nutrition provides the energy required to carry out various living activities. Organisms are classed as autotrophs or heterotrophs based on their mode of nourishment.

**A. Autotrophic Nutrition:**

When organisms obtain their organic components from inorganic sources, they are said to be autotrophic. Carbon dioxide and nitrates are the primary sources of carbon and nitrogen, respectively. All green plants are autotrophic, meaning they use light as a source of energy to produce food via photosynthesis.

$$6CO_2 + 12H_2O \xrightarrow{Chlorophyll Sunlight} C_6H_{12}O_6 + 6H_2O + 6O_2$$

**During this process, the following things happen.**

**i.** Light energy absorption by chlorophyll.

**ii.** Light energy is converted to chemical energy, and water molecules are divided into hydrogen and oxygen.

**iii.** Carbon dioxide is converted to carbohydrates.

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**B. Heterotrophic Nutrition: Heterotrophs are entirely reliant on autotrophs for their food.**

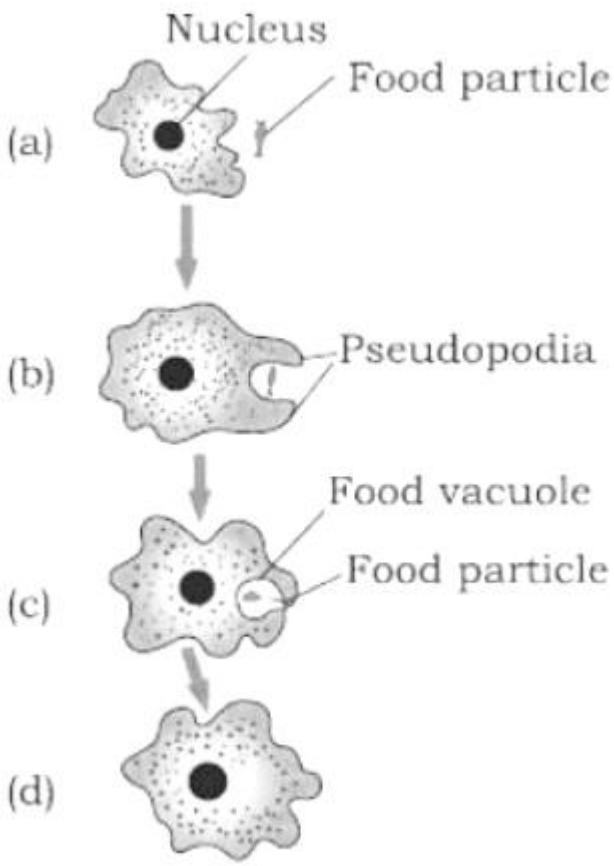
**Heterotrophic nutrition is divided into three categories:**

1. **Holozoic Nutrition:** Complex foods are ingested through a specialised digestive system and broken down into little pieces so that they can be absorbed. For example, Amoeba and Humans.

2. **Saprophytic Nutrition:** It refers to the feeding of organisms on the dead organic remnants of other creatures. Fungi, for example, enjoy bread moulds, yeast, and mushrooms.

3. **Parasitic Nutrition:** Parasites feed on other living beings (the host), with no benefit to the host. Cuscuta, ticks, lice, leeches, and tapeworms are examples.

**How Do Organisms Obtain their Nutrition?**




Food can be taken in by the entire surface of single-celled organisms. For example, an amoeba takes in food by building a food-vacuole out of transient finger-like extensions of the cell surface that fuse over the food particle. Complex compounds are broken down inside the food vacuole into simpler ones, which eventually diffuse into the cytoplasm. The remaining undigested material is pushed to the cell's surface and discarded.

- **Human Nutrition:** Food digestion takes place in the alimentary canal, which is made up of several organs and glands.
- The remaining undigested material is moved to the surface of the cell and thrown out.

- Food is chewed into minute particles in the mouth and combined with saliva, which contains amylase for starch digestion.
- Food goes through the pharynx and oesophagus on its way to the stomach when swallowed. Pepsin (a protein-digesting enzyme),
- HCl and mucus are found in gastric juice.
- The hydrochloric acid generates an acidic environment for the enzyme pepsin to work in. Under normal circumstances, mucus protects the stomach's inner lining from the acid's activity.
- The meal now enters the small intestine from the stomach. The small intestine is where carbs, proteins, and lipids are completely digested.
- Bile is secreted by the liver, which emulsifies fat.
- Pancreatic juice, which contains the enzymes amylase, trypsin, and lipase for digesting starch, proteins, and fats, is secreted by the pancreas.
- Carbohydrates, proteins, and fats are entirely digested in the small intestine, resulting in glucose, amino acids, fatty acids, and glycerol.
- The small intestine villi absorb the digested food and deliver it to all of the body's cells.
- The undigested food is transferred to the large intestine, where it is absorbed by additional villi. The anus is used to eliminate the rest of the material from the body.

**Nutrition in Paramoecium**

- Paramoecium attains nutrition through the process of holozoic nutrition, where the organism ingests liquid or solid food particles inside its body.
- The cilia present on the Paramoecium help them to engulf the food particles through the oral groove.

- The digested food passes through the cytoplasm by the process of cyclosis and reaches the food vacuoles, which is then absorbed by the cytoplasm.
- The undigested food is later, taken out through the tiny pore known as cytopyge or anal pore.

**Respiration:**

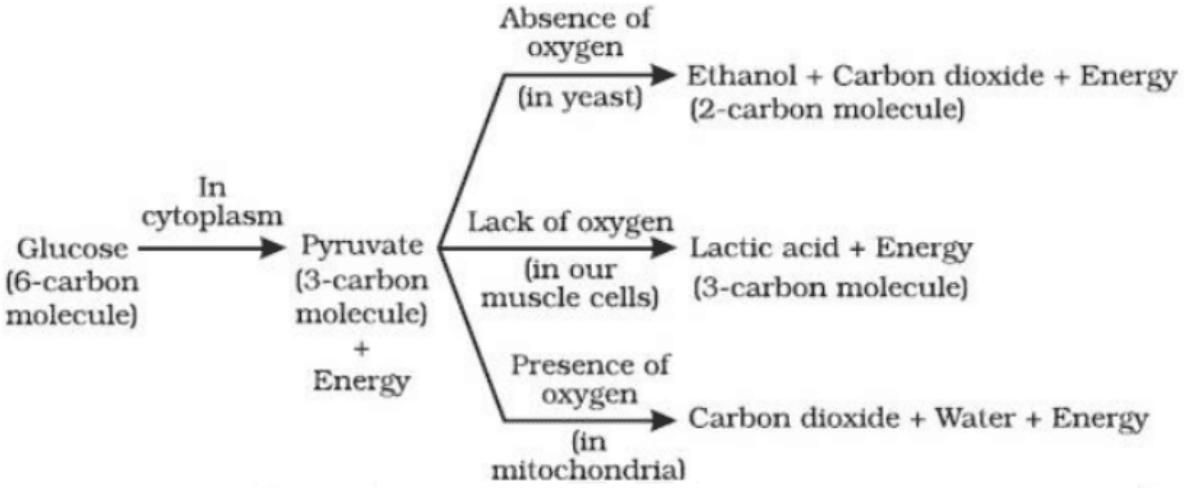
The digested food items are broken down during breathing to release energy in the form of ATP.

**Respiration can be one of two forms, depending on the amount of oxygen required:**

i. **Aerobic respiration:** This occurs when there is air present (oxygen).

ii. **Anaerobic respiration:** This occurs when there is no (air) oxygen present.

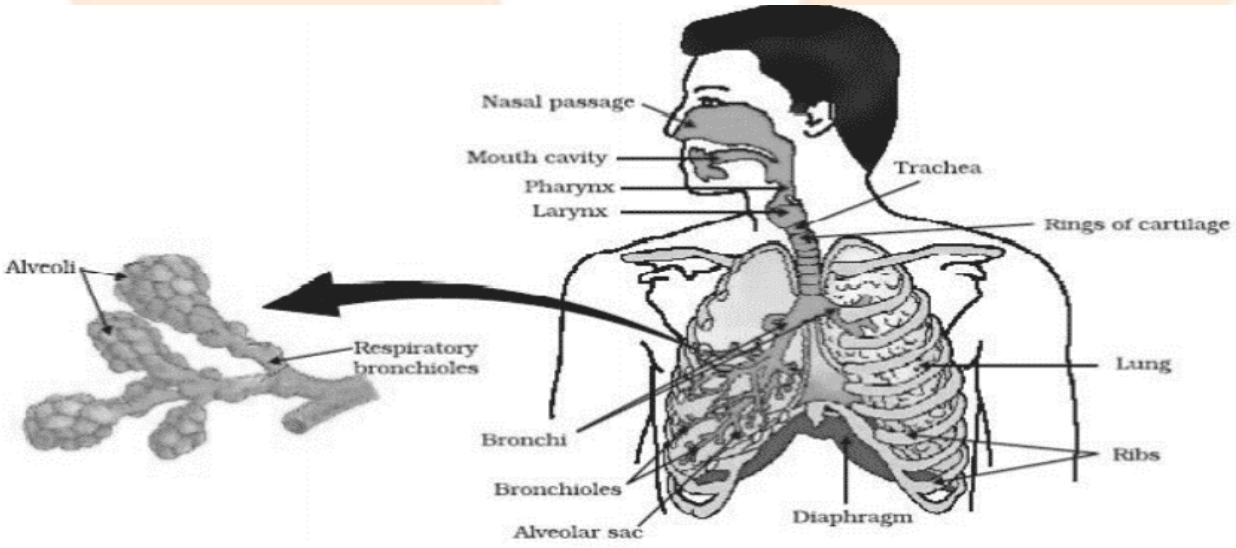
- The breakdown of glucose, a six-carbon molecule, into pyruvate, a three-carbon molecule, is the first step in all cases. The cytoplasm was involved in this process. Pyruvate can also be converted to ethanol and carbon dioxide. During fermentation, yeast undergoes this process. Anaerobic respiration is the name given to a process that occurs without the presence of oxygen (air). In the mitochondria, pyruvate is broken down utilising oxygen.
- When compared to anaerobic respiration, aerobic respiration releases a lot of energy.
- When there is a shortage of oxygen in our muscle cells, pyruvate is transformed into lactic acid. Cramping occurs when lactic acid builds up in our muscles as a result of an abrupt activity.




Aquatic species use oxygen dissolved in water for respiration, whereas terrestrial organisms use air oxygen.

In humans, inhalation of air occurs through the following pathway: Nostrils

→Nasal passage→ Pharynx → Larynx → Trachea→ Bronchus → Bronchiole→ Alveolus



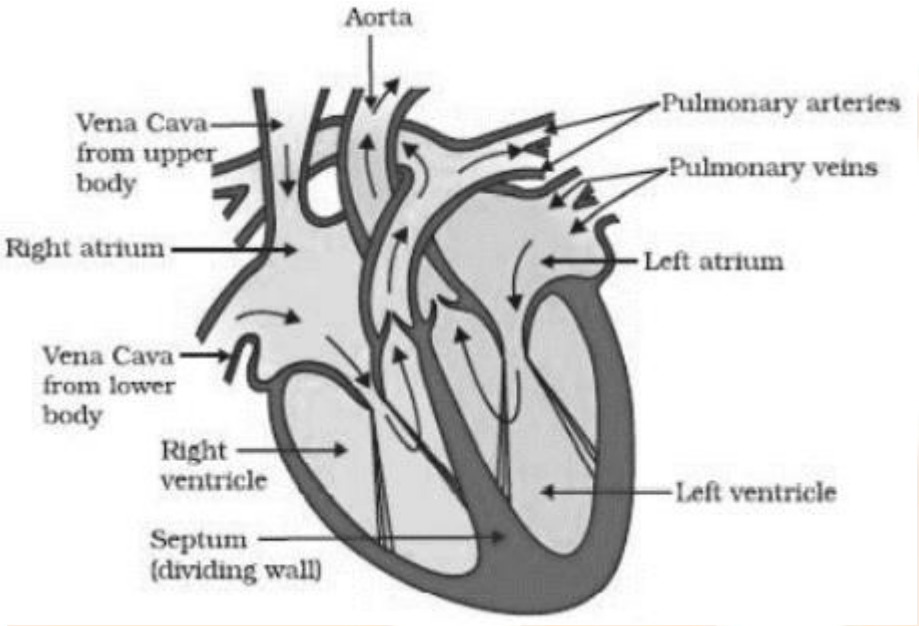
- Air enters the body via the nostrils in humans. Fine hairs line the channel, filtering the air that passes through the nose. Mucus lines the channel, assisting in the process. The air then enters the lungs via the throat. The neck is lined with cartilage rings. They keep the airway from collapsing.
- The channel within the lungs divides into smaller and smaller tubes that eventually finish in balloon-line formations known as alveoli.
- The alveoli of the lungs are densely packed with blood and serve as locations for gas exchange
- ($O_2$ *and* $CO_2$O and CO) between the blood and the atmosphere.
- The blood transports carbon dioxide from the rest of the body to the alveoli, while blood in the alveolar blood vessels absorbs oxygen from the alveolar air and transports it to all of the body's cells. During the breathing cycle, when air is inhaled and exhaled, the lungs always have a residual volume of air so that oxygen may be absorbed and carbon dioxide can be exhaled in adequate time.
- The respiratory pigment haemoglobin transports oxygen from the lungs to the body's various tissues in humans. The red blood cells contain this pigment.

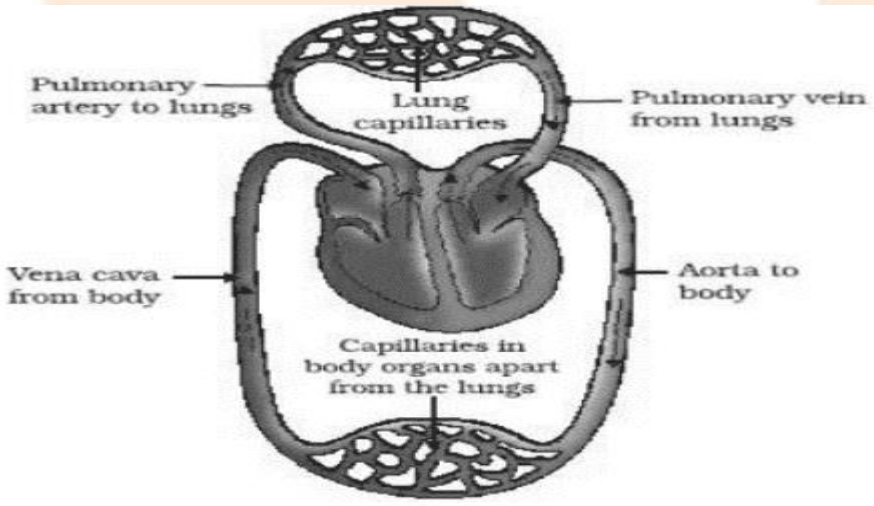
**Transportation:**

- **Transportation in Human Beings:**

The heart, blood, and blood arteries make up the circulatory system, which transports various materials throughout the body.

**The Heart:**


Aorta
Vena Cava from upper body
Pulmonary arteries
Pulmonary veins
Right atrium
Left atrium
Vena Cava from lower body
Right ventricle
Left ventricle
Septum (dividing wall)


Pulmonary artery to lungs
Lung capillaries
Pulmonary vein from lungs
Vena cava from body
Aorta to body
Capillaries in body organs apart from the lungs

- The human heart is divided into four chambers: two atria (right and left) and two ventricles (right and left) (right and left). These chambers prevent oxygen-rich blood from mixing with carbon dioxide-rich blood. Deoxygenated blood is delivered to the right half of the heart, while oxygenated blood is delivered to the left.
- The carbon dioxide-rich blood must reach the lungs in order for the carbon dioxide to be eliminated, and the oxygenated blood must return to the heart. The oxygen-rich blood is subsequently circulated throughout the body.
- The walls of the ventricles are significantly thicker than the walls of the atrium.
- Humans have twofold circulation, which means blood passes through the heart twice and oxygenated and deoxygenated blood are separated completely.
- Arteries transport blood from the heart to various regions of the body, while veins return it to the heart. Thin capillaries connect arteries to veins, allowing materials to be exchanged between the blood and cells.
- Platelet cells are found in blood and circulate throughout the body, preventing blood loss at the site of damage.
- Lymph has a role in transportation as well. It's similar to blood plasma, but it's colourless and has less protein. From the intercellular gaps, it empties into lymphatic capillaries, which connect to major lymph arteries, which then open into bigger veins. It transports fat from the intestine that has been digested and absorbed, as well as excess fluid from extracellular space back into the circulation.

**Transportation in plants:**

- Plant transport systems will carry raw materials from roots and energy stores from leaves. These two paths are made up of conducting tubes that are set up separately. The xylem, for example, transports water and minerals from the soil. The phloem, on the other hand, carries photosynthetic products from the leaves to other sections of the plant.

- The xylem tissue of roots stems, and leaves (tracheids and vessels) are interwoven to form a continuous system of water transmitting channels that reaches all parts of the plant. Water is driven into the xylem cells of the roots as a result of suction pressure created by transpiration. Then, through the interconnecting water-carrying channels, water moves steadily from the root xylem to all sections of the plant.

- Transpiration is the loss of water from the plant's aerial portions in the form of vapour. As a result, it aids in the absorption and upward transfer of dissolved minerals in water from the roots to the leaves. It also controls the temperature.

- Translocation happens in phloem and is the movement of soluble photosynthetic products. Amino acids and other chemicals are transported via it. Food and other substances are transported in the sieve tubes in both upward and downward directions with the help of nearby partner cells.

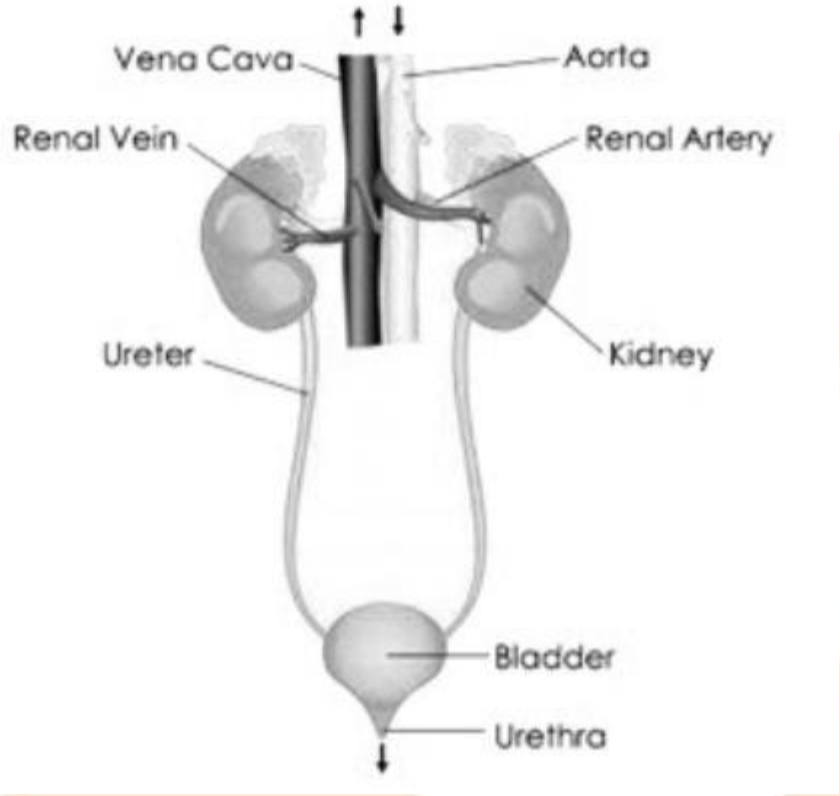
- Energy is used to facilitate translocation in the phloem. Sucrose is transported into phloem tissue with the help of ATP energy. This causes the tissue's osmotic pressure to rise, allowing water to enter. This is the pressure. This allows the phloem to transfer material in accordance with the needs of the plant. For example, in the spring, sugar stored in root or stem tissue would be transported to the buds which need energy to grow.

**Excretion:**

The toxic metabolic nitrogenous wastes produced are excreted from the body during excretion.

**Excretion in Human Beings:**

- The excretory system in humans consists of two kidneys, two ureters, the urine bladder, and the urethra. The abdomen has two kidneys, one on each side of the backbone. The ureters transport urine from the kidneys to the urinary bladder, where it is held until it is discharged through the urethra.




- Nephrons are basic filtration units found in great numbers in each kidney. As the urine passes along the tube, several substances in the initial filtrate, such as glucose, amino acids, salts, and a significant amount of water, are selectively reabsorbed. The amount of water reabsorbed is determined by the amount of excess water in the body as well as the amount of dissolved waste to be expelled. The urine produced by each kidney eventually enters the ureter, a lengthy tube that connects the kidneys to the urinary bladder until the inflated bladder causes the urge to send it out through the urethra. Because the bladder is muscular, it is controlled by the neurological system. As a result, we are able to manage our urge to urinate.

**Excretion in plants:**

- Plants do not have an excretory system and excrete through a variety of mechanisms, including transpiration, waste release into the surrounding soil, leaf loss, and waste storage in cell vacuoles. Other waste materials are retained in ancient xylem as resins and gums.